

KARRATHA KING BAY, Australia

WMO: 953080

Lat: 20.617S

Lon: 116.750E

Elev: 52

StdP: 14.67

Time Zone: 8.00 (E08)

Period: 91-01

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	61.3	62.7	26.3	20.3	74.7	30.5	24.7	73.5	35.2	68.5	32.6	67.5	17.9	110	0.587

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	11.7	97.8	74.7	95.3	75.6	93.5	76.3	85.3	90.1	83.5	89.1	82.2	88.4	15.4	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
84.3	180.3	88.6	82.0	166.9	87.5	80.5	158.9	86.5	50.0	90.4	47.7	89.6	46.1	88.3	93.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 32.6	29.3	26.0	56.2	106.5	3.6	4.5	53.6	109.8	51.5	112.4	49.5	114.9	46.8	118.2		
(5)			45.4	88.6	1.7	4.1	44.2	91.5	43.2	93.9	42.2	96.2	41.0	99.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	79.6	85.4	85.9	86.8	84.5	77.4	72.0	70.8	73.0	74.9	79.5	81.3	84.0
	DBStd	6.66	3.14	3.35	3.41	3.15	3.76	4.17	3.01	3.27	3.26	4.36	4.15	3.79
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	6	0	0	0	0	0	2	3	1	0	0	0	0
	CDD50	10795	1097	1005	1140	1034	850	660	643	712	747	915	938	1054
	CDD65	5325	632	585	675	584	385	212	181	247	297	450	488	589
	CDH74	54845	8165	7768	9078	7071	2900	820	454	1010	1567	3978	4964	7070
	CDH80	22666	3816	3828	4717	3112	673	86	20	129	271	1285	1749	2980
Wind	WSAvg	14.3	15.3	15.2	13.6	12.1	13.3	14.2	13.6	13.0	13.4	14.8	16.4	16.3
Precipitation	PrecAvg	11.8	2.0	3.4	1.7	0.8	1.2	1.5	0.4	0.1	0.0	0.0	0.1	0.5
	PrecMax	29.8	9.8	14.1	8.1	4.9	6.4	6.5	2.8	0.9	0.7	0.6	1.5	4.2
	PrecMin	2.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	7.2	2.7	4.1	2.4	1.2	1.6	1.8	0.6	0.2	0.2	0.1	0.3	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	99.1	100.1	101.3	96.9	91.3	86.2	82.3	86.4	90.9	97.7	97.8	100.6
		MCWB	75.2	77.2	72.6	71.2	68.4	69.0	61.5	61.8	67.0	68.0	69.5	71.3
	2%	DB	94.5	95.7	97.1	94.6	88.1	82.1	79.9	83.1	86.0	91.6	93.1	94.5
		MCWB	80.3	80.0	74.5	71.0	69.4	67.3	62.2	61.8	67.8	68.5	71.2	77.5
	5%	DB	92.4	93.3	94.6	92.6	85.8	80.2	78.2	81.2	83.2	89.0	90.2	92.0
		MCWB	80.1	79.9	76.7	71.6	69.1	66.9	63.0	62.5	66.8	68.9	72.7	78.0
	10%	DB	90.4	91.4	92.6	90.7	83.7	78.4	76.6	79.3	81.0	86.3	87.8	89.7
		MCWB	79.8	79.8	77.3	72.4	69.3	66.5	62.7	62.8	66.7	69.9	72.8	78.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	87.8	87.4	87.5	85.3	78.3	75.6	74.2	72.9	76.2	78.1	82.3	85.6
		MCDB	92.7	92.4	91.6	90.0	83.9	79.6	76.5	75.5	83.9	84.9	88.0	90.2
	2%	WB	85.1	84.6	82.9	80.8	76.5	73.2	70.9	70.8	72.8	76.5	79.7	83.4
		MCDB	89.4	90.1	90.4	88.1	81.6	77.4	75.0	75.6	80.2	82.9	85.2	88.3
	5%	WB	83.3	82.8	81.7	79.0	75.1	71.9	68.8	69.7	71.5	75.2	78.1	81.9
		MCDB	88.2	89.5	89.2	85.8	80.8	76.9	74.4	75.2	78.7	82.3	84.2	87.1
	10%	WB	81.7	81.6	80.6	77.7	73.6	70.1	67.0	68.3	70.1	73.9	76.9	80.4
		MCDB	87.4	88.3	88.0	85.0	79.8	76.3	73.2	74.8	77.5	81.5	83.2	86.4
Mean Daily Temperature Range	5% DB	MDBR	9.0	9.4	11.7	12.6	11.0	10.3	11.7	12.7	12.1	12.3	11.3	9.6
		MCDBR	11.3	12.7	14.2	15.1	13.0	11.7	13.2	14.7	14.6	16.0	15.0	13.0
		MCWBR	7.1	8.1	9.4	10.6	8.7	7.9	9.5	10.0	10.8	10.6	10.0	8.0
	5% WB	MCDBR	9.5	10.4	11.5	12.4	10.7	8.9	10.8	11.4	12.3	12.5	11.7	9.9
		MCWBR	6.7	7.2	8.6	9.6	7.8	8.1	9.6	8.8	9.1	8.9	8.8	7.0
Clear-Sky Solar Irradiance	taub	0.402	0.397	0.378	0.353	0.335	0.323	0.316	0.314	0.333	0.357	0.359	0.378	
	taud	2.517	2.546	2.589	2.615	2.600	2.622	2.615	2.582	2.528	2.487	2.519	2.539	
	Ebn at Noon	299	297	295	290	284	282	287	298	304	306	310	307	
	Edh at Noon	36	35	32	29	27	25	26	29	33	36	36	35	
All-Sky Solar Radiation	RadAvg	2368	2267	2023	1763	1471	1334	1449	1789	2114	2390	2548	2544	
	RadStd	144	155	144	95	113	94	93	50	80	50	58	98	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	+0.83	N/A	N/A	N/A	+0.86	+1.01	N/A	N/A	N/A	+284

Nomenclature: See separate page